

Qbit ULL SmartNic

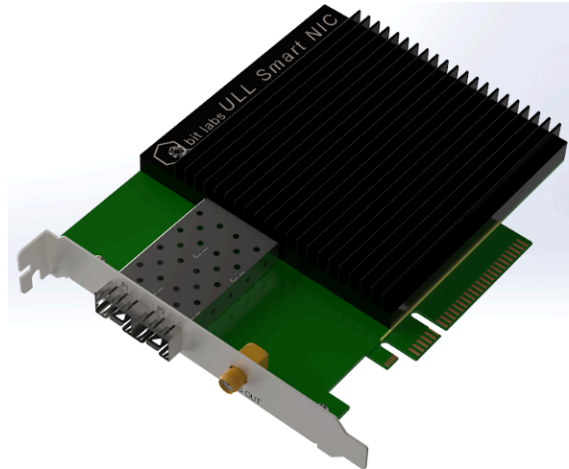
The Qbit NextGen SmartNIC is a high-performance network acceleration solution designed for ultra-low latency and high-speed Ethernet applications. Engineered with a focus on scalability and efficiency, it addresses the stringent requirements of latency-sensitive environments by combining advanced hardware design, optimized gate utilization, and streamlined data-path architecture to deliver consistent, high-speed performance.

It supports full line-rate throughput across packet sizes from 64-byte frames to jumbo packets, ensuring reliable, loss-free data handling even under mixed traffic conditions.

Introduction

The Qbit NextGen SmartNIC is a high-performance, FPGA-based network interface solution designed to deliver ultra-low latency and high-speed Ethernet connectivity for demanding, latency-sensitive environments. It is engineered using advanced hardware design techniques that optimize gate utilization and data-path efficiency, enabling deterministic performance and scalable processing capabilities.

The SmartNIC sustains full line-rate throughput across packet sizes ranging from standard 64-byte frames to jumbo packets, ensuring seamless, loss-free data handling even under mixed traffic conditions. Its flexible FPGA architecture supports hardware acceleration, custom logic integration, and efficient packet processing features such as classification, filtering, and flow steering. It also provides precision timing support through hardware timestamping and synchronization features, making it ideal for applications such as high-frequency trading, high-performance computing, and real-time data processing.



Architecture

The SmartNIC architecture is designed around an FPGA-based high-speed data path that connects dual 10/25G Ethernet ports to a host system through a PCIe interface. Network traffic enters through the ports and is first handled by the MAC/PCS block, which performs physical layer functions such as encoding/decoding and Ethernet frame processing. The data then moves into a programmable FPGA pipeline where several hardware modules carry out tasks such as custom packet filtering, pattern matching, offload processing, and transmit frame formatting. This hardware-level processing allows packets to be analyzed, modified, or forwarded in real time without relying heavily on the host CPU.

Once processing is complete, the SmartNIC transfers data to the host via a PCIe Gen4 x4 connection, enabling high-speed communication with system memory and applications. An API layer on the host side allows software to configure, control, and interact with the SmartNIC functions. In addition to the main data path, the architecture includes support interfaces such as USB-JTAG for configuration and debugging, as well as a 1-PPS input/output for precise time synchronization. Together, these components provide a flexible and efficient platform for accelerating networking workloads, reducing CPU overhead, and supporting time-sensitive applications.

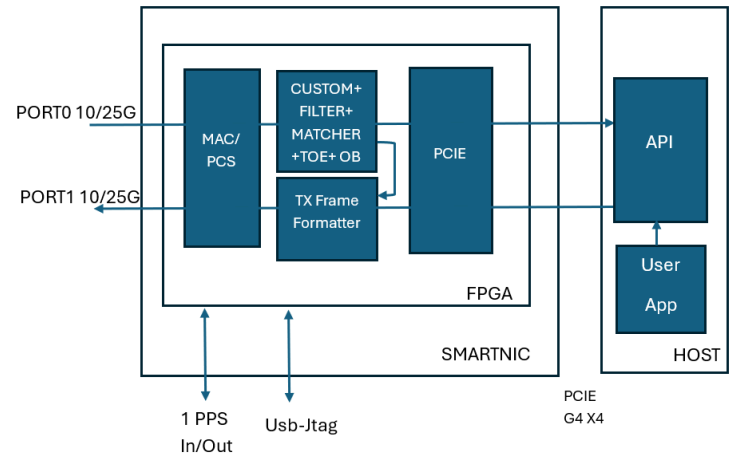
Why Choose Us?

-  Rapid integration, faster time-to-market
-  Clean, verified, documented RTL
-  Direct access to experts



Features

- Ultra-low latency with ~600 ns tick-to-trade and <500 ns through dedicated FPGA data paths.
- Full line-rate processing from 64-byte packets up to jumbo frames without packet loss.
- Support for 10G/25G Ethernet with multiple SFP+/SFP28 ports.
- High-speed PCIe Gen4 x4/x8 host interface for fast data transfer.
- Precision hardware timestamping (~1.6 ns resolution).
- IEEE-1588 PTP synchronization and 1 PPS input/output support.
- Advanced packet classification and flow steering across up to 32 queues.
- FPGA-based architecture allowing custom logic addition through FDK.
- Socket acceleration library support for existing applications.
- Kernel mode operation and low-latency raw Ethernet mode.
- Real-time traffic capture at live data rates for analysis.
- Performance monitoring and error counter support.
- Built-in MAC filtering, multicast filtering, and VLAN tagging support.
- TCP/UDP checksum offload and TCP segmentation support.
- RSS support for distributing packets across multiple queues.
- Zero-copy data transfer and multicast acceleration.
- Support for IPv4/IPv6, ARP, ICMP, and IGMP protocols.
- Operating temperature range from 0 °C to 55 °C.



Specifications

Hardware Specifications	
Category	Phase Noise
PHY Interfaces	• 2/4 SFP+ / SFP28 Ports
Port Speed	• All ports support up to 25G
Host Interface	• PCIe Gen4 x4 / x8 interface to host
Debug / Programming	• USB to JTAG interface



Performance & Acceleration Features	
Features	Details
Tick-to-Trade Latency	• ~600 ns (host application path)
FPGA Path Latency	• < 500 ns
Packet Processing	• Classification & flow steering up to 32 queues
Socket Acceleration	• NextGen socket acceleration library
NIC Modes	• Kernel mode & low-latency raw Ethernet mode
Customization	• Custom FPGA logic through FDK
Data Capture	• Live data rate capture to disk
Trading Support	• Pre-trade & post-trade checks
Monitoring	• Performance and error counters
NIC Functions (Protocol Stack Support)	
Layer	Functions
PHY (L1)	• 10G / 25G, Auto-negotiation, FEC
DLL (L2)	• Ethernet framing, MAC filtering, multicast filtering • Flow control (pause frames), VLAN tagging/filtering • CRC error check, hardware timestamping
Network (L3)	• IPv4, IPv6, ARP, ICMP, IGMP
Transport (L4)	• TCP/UDP checksum offload • TCP segmentation • RSS – packet distribution across queues • Zero-copy processing • Multicast acceleration
Synchronization	• PTP IEEE-1588, 1 PPS



Supported Exchange Protocols	
Category	Protocols
Market Data	• NSE NNF / NSE MTBT
Financial Messaging	• FIX
Low-Latency Trading	• OUCH / ITCH
Data Compression	• FAST
Environmental Specifications	
Parameter	Details
Operating Temperature Range	• 0 °C to 55 °C
Compliance / Timing Standards Supported	
Category	Standard / Support
Time Synchronization	• IEEE-1588 PTP support
External Timing Interface	• 1 PPS input / output
Precision Timing	• Hardware timestamping (1.6 ns resolution)

Get in touch with us

Delivering Advanced Solutions for
Next-Gen Networking



104, FF, Iris Tech Park, Sector 48
Sohna Road, Gurugram, 122018



+91 124 4048731



sales@qbitlabs.co.in